

Seongmoon Jeong

PHD CANDIDATE

IRIS Lab., Sungkyunkwan University, Suwon, Republic of Korea

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027)

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2013 - Feb. 2020

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Machine-Oriented Image Compression:** Codec-transferable and test-time adaptation across downstream vision tasks [**C4**, **C5**]
- Task-Aware Preprocessing and JPEG Coding:** Adaptive downscaling, feature modulation, and region-based JPEG optimization [**C1**, **C3**, **J1**]
- RAW-Domain Machine Vision:** Foundation-model-guided RGB-to-RAW generation for object detection [**C6**]

Project Experience

Video Coding for Machines

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2023 - 2024

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2022

Selected Publications

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

IEEE CVPR

Mar. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

IEEE AICAS

Feb. 2024

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko

IEEE ICME Workshop

Apr. 2023

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko

Pattern Recognition Letters

Nov. 2022

Standardization Activities

[VCM] Neural network based post-filter for VCM ISO/IEC JTC1 / SC29 / WG4 M71305	<i>Geneva</i> <i>Jan. 2025</i>
[VCM] Neural network based pre-filter for VCM ISO/IEC JTC1 / SC29 / WG4 M71304	<i>Geneva</i> <i>Jan. 2025</i>
[VCM] Learned joint filter network for VCM ISO/IEC JTC1 / SC29 / WG4 M69990	<i>Kemer</i> <i>Nov. 2024</i>
[VCM] Learned pre-filter network for VCM ISO/IEC JTC1 / SC29 / WG4 M69085	<i>Sapporo</i> <i>Jan. 2024</i>

Honors & Awards

2024	3rd Place , DAC System Design Contest - GPU Track	<i>San Francisco, USA</i>
2020	2nd Place , Artificial Intelligence Grand Challenge, Ministry of Science and ICT	<i>Korea</i>
2019	Honorable Mention , Big Data Center Open Innovation Challenge	<i>Seongnam, Korea</i>
2019	Outstanding Work Award , Capstone Design Project, SKKU	<i>Suwon, Korea</i>
2018	Dean's List Award , Fall semester, SKKU	<i>Suwon, Korea</i>
2017	Dean's List Award , Fall semester, SKKU	<i>Suwon, Korea</i>

Full Publications

INTERNATIONAL CONFERENCE

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko	<i>IEEE AVSS</i> <i>Sep. 2026</i>
C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko	<i>IEEE AVSS</i> <i>Sep. 2026</i>
C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko	<i>IEEE CVPR</i> <i>Mar. 2025</i>
C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko	<i>IEEE AICAS</i> <i>Feb. 2024</i>
C2 Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko	<i>IEEE ICCAD</i> <i>Jul. 2023</i>
C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko	<i>IEEE ICME Workshop</i> <i>Apr. 2023</i>

INTERNATIONAL JOURNAL

J2 KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko	<i>IEEE TCAS-I</i> <i>Feb. 2024</i>
J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko	<i>Pattern Recognition Letters</i> <i>Nov. 2022</i>

Student Mentoring

Hangyul Choi M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS	<i>2023 - 2024</i>
• Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression	
AUGUST 15, 2026	SEONGMOON JEONG · CURRICULUM VITAE

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko

ASSOCIATE PROFESSOR

- iris.skku.edu
- jhko@skku.edu

Ph.D. advisor

Dept. of ECE, SKKU, Suwon, Korea